

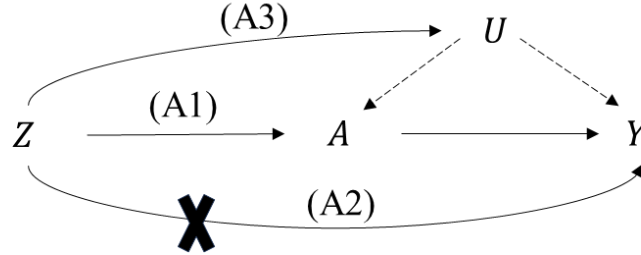
Causal Inference – Final Exam

Exam Date: Dec 12, 2025

Exam Duration: 15:00 ~ 18:00 (3 hours)

Dept / Year: 統計三、統計四

1. A variable is considered a valid instrument if it satisfies three main conditions, labeled (A1) – (A3) in the diagram below:



- (10%) State and briefly explain conditions (A1) – (A3).
 - (10%) Are conditions (A1) – (A3) sufficient for identification of IV estimators? If not, what additional conditions are required for identification?
2. Consider the following regression model

$$Y = \alpha_0 + \alpha_1 A + \varepsilon$$

Assume the regressor A is endogenous. To address this, the research employs three instrumental variables: Z_1, Z_2 and Z_3 . After estimating the model using **Two Stage Least Square (2SLS)**, the researcher obtains the residuals, $\hat{\varepsilon}$, and runs an auxiliary regression of these residuals on the instruments.

$$\hat{\varepsilon} \sim Z_1 + Z_2 + Z_3$$

The R-squared from this auxiliary regression is $R^2 = 0.012$, and the sample size is $n = 500$.

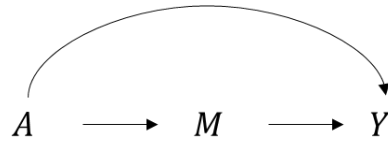
- (5%) Is the IV model overidentified? Briefly explain your answer.
 - (10%) State the null and alternative hypotheses, as well as the decision rule, for the Sargan Test.
 - (10%) Calculate the Sargan test statistics. At 5% significance level, determine whether there is sufficient evidence to reject the null hypothesis regarding the validity of the instruments.
- (Hint: $\chi^2_{(1),0.95} = 3.84, \chi^2_{(1),0.975} = 5.02, \chi^2_{(2),0.95} = 5.99, \chi^2_{(2),0.975} = 7.38$)
3. Consider an endogenous regressor A and a single instrument Z . Let $\hat{\beta}_{IV}$ and $\hat{\beta}_{OLS}$ denote the IV and OLS estimators, with asymptotic variance $\text{Var}(\hat{\beta}_{IV})$ and $\text{Var}(\hat{\beta}_{OLS})$, respectively. You may use the following approximation identity:

$$\frac{\text{Var}(\hat{\beta}_{IV})}{\text{Var}(\hat{\beta}_{OLS})} \approx \frac{1}{R^2_{A \sim Z}}$$

where $R^2_{A \sim Z} = 0.01$ is the 1st stage R squared. Suppose $\hat{\beta}_{IV} \approx \hat{\beta}_{OLS} = 7$ and the standard error $\text{se}(\hat{\beta}_{IV}) = 2$.

- (5%) Explain the importance of having a strong instrument. Demonstrate this using the identity provided above.
 - (5%) Compute the implied OLS standard error $\text{se}(\hat{\beta}_{OLS})$.
 - (10%) Construct the confidence intervals for both IV and OLS estimate by using ± 2 times their respective standard errors.
4. Consider a causal framework where an exposure A affects an outcome Y , potentially through a

mediator M . The relationships are estimated using the following two regression models:



$$Y = 2 + 1.2A + \varepsilon$$

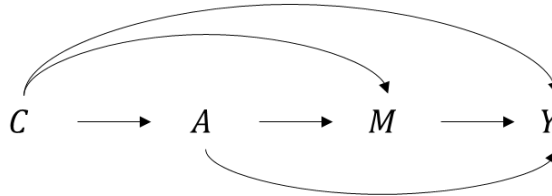
$$Y = 5 + 0.7A + 3M + v$$

- a. (5%) Using the **Difference Method**, calculate the **NIE** and **TE**.
 - b. (10%) Calculate the **Proportion Mediated (PM)** and determine which pathway (direct or indirect) is dominant in influencing the outcome
5. A manufacturing company is investigating whether improving Soundproofing Quality (A) improves Worker Productivity (Y), and whether this relationship is mediated by Worker Concentration (M). The regression results are provided below:

Effect	Path	Coefficient	Standard Error
α	$A \rightarrow M$	0.45	0.15
β	$M \rightarrow Y$ (controlling A)	0.2	0.12

Assume the sample size is sufficiently large such that the critical value for two-tailed t-test at the 5% significance level is 1.96.

- a. (5%) Calculate the t-statistics for coefficients α and β .
 - b. (10%) State the hypotheses and the decision rule for the **Intersection-Union Test** in its general form (i.e., for a set of hypotheses H_{0i} vs H_{1i}).
 - c. (10%) Using the **Joint Significant Test (Intersection Union Test)** determines whether the indirect effect $\alpha\beta$ is statistically significant at the 5% level. Justify your conclusion.
6. Mediation analysis addresses questions regarding mechanisms M through which the causal effect of A on Y operates. A series of exchangeability assumptions are crucial to avoid confounding.



- a. (15%) Specify the four key assumptions of exchangeability (No unmeasured confounding).
 - b. (10%) Describe the scenario in which the cross-world exchangeability assumption is violated. Provide an explanation using the **Non-Parametric Structural Equation Models (NPSEM)**.
7. Consider the mediation formula defined by

$$Q(a, a^*) = \sum_c \left\{ \int_m E(Y|M = m, A = a, C = c) f_{M|A,C}(m|a^*, c) dm \right\} P(C = c)$$

Given that Y is a rare binary outcome, assume the following log-linear model

$$\log\{E(Y|A = a, M = m, C = c)\} = \beta_0 + \beta_1 a + \beta_2 m + \theta_c c$$

Furthermore, assume the mediator M follows a conditional normal distribution given by

$$M|A = a^*, C = c \sim N(\alpha_0 + \alpha_1 a^* + \alpha_c c, 2[1 + a^*]\sigma^2)$$

- a. (10%) Compute $Q(1,1), Q(1,0), Q(0,1)$ and $Q(0,0)$.
- b. (10%) Compute **NIE** and **NDE** in **Risk Ratio** scale.

(Hint: Moment Generating function)